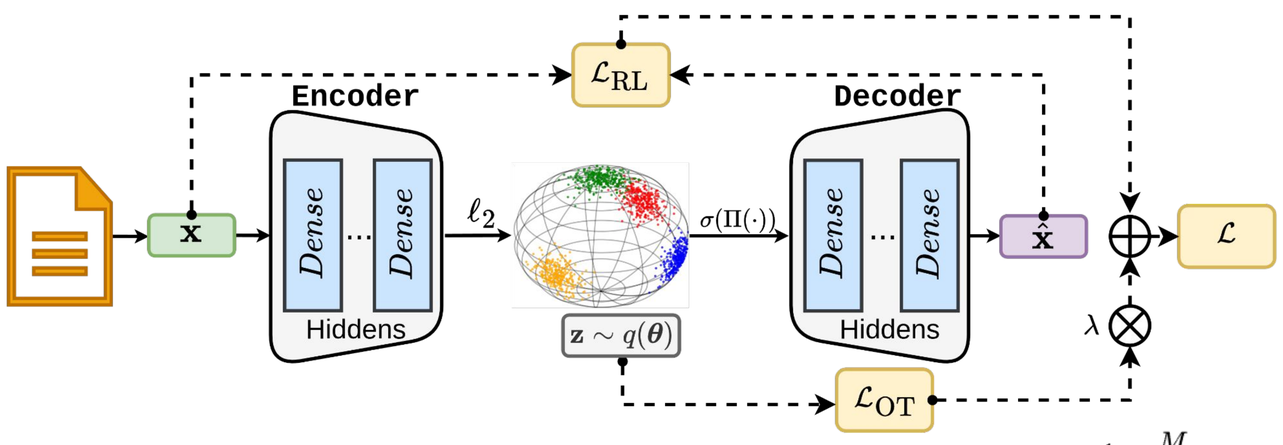
S2WTM: Spherical Sliced-Wasserstein Autoencoder for Topic Modeling

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S2WTM is a Spherical Topic Model [1] that represents a document with directional distributions on the unit hypersphere. It uses a hyperspherical prior and aligns the posterior—also on the hypersphere—via the Spherical Sliced-Wasserstein distance, avoiding posterior collapse and ensuring regularization over the same space.

Proposed Framework



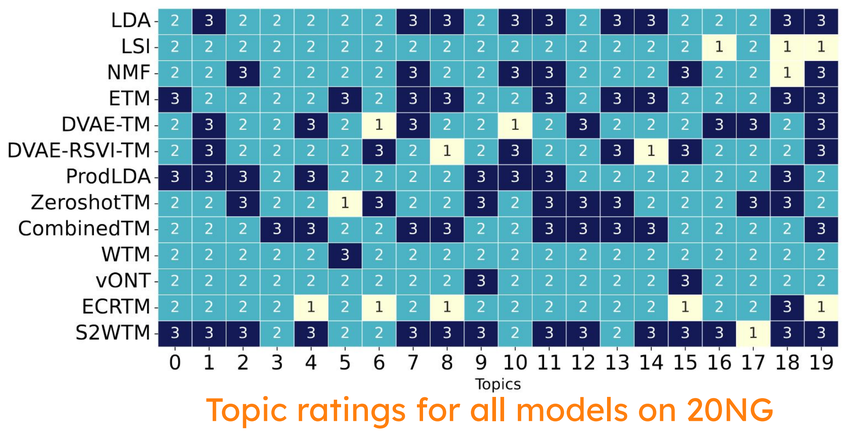
$$\mathcal{L} = \inf_{q(\theta|\mathbf{x})} \mathbb{E}_{p(\mathbf{x})}\mathbb{E}_{q(\theta|\mathbf{x})}\left[c(\mathbf{x},\hat{\mathbf{x}})\right] + \lambda \cdot \mathrm{SSW}_2^2\left(q_\theta, p_\theta\right), \text{ where } \mathrm{SSW}_2^2\left(q_\theta, p_\theta\right) \approx \frac{1}{M}\sum_{i=1}^{M} W_2^2\left(R_i q_\theta, R_i p_\theta\right)$$

Quantitative Analysis

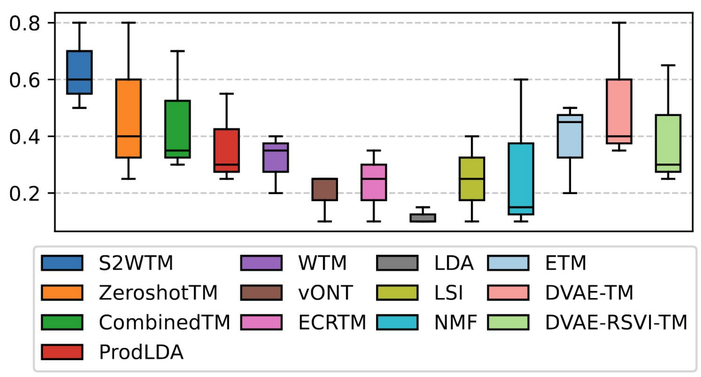


Topic coherence (NPMI and CV) scores for few selected topic models.

LLM-based Topic Quality Assessment



Topic ratings for all models on 20NG



Intruder detection accuracy (3 runs) on 20NG

LLM-based Topic Quality Assessment serves as a proxy for human interpretability, addressing the limitations of automated topic quality evaluations.

Qualitative Evaluation

Model	Topics
WTM	**game, team, player, play, year**
	drive, *work, price, good, sell*
	key, government, system, encryption, *make*
vONT	**game, year, good, time,** *make*
	drive, *problem*, **card, run, system**
	key, chip, encryption, government, bit
S2WTM	**game, team, win, score, player**
	drive, card, scsi, ide, bus
	encryption, secure, chip, encrypt, phone

Three topics sports, hardware, and encryption/security from the 20NG dataset.

Beyond Euclidean Assumptions

Dataset	NPMI		CV	
	Euclidean	Spherical	Euclidean	Spherical
20NG	0.108	0.167	0.623	0.723
BBC	0.081	0.252	0.592	0.863
M10	0.038	0.101	0.494	0.492
SS	0.024	0.146	0.549	0.683
Pascal	0.007	0.045	0.425	0.572
Bio	0.094	0.191	0.572	0.663
DBLP	0.004	0.133	0.369	0.558

Topic coherence (NPMI, CV) comparison between Euclidean and Hyperspherical latent spaces

Curse of dimensionality

As $d \to \infty$, for $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$, we have:

$$\|\mathbf{z}\| \xrightarrow{p} \sqrt{d} \;\Rightarrow\; \mathcal{N}(\mathbf{0}, \mathbf{I}_d) \approx \mathcal{U}(\mathbb{S}^{d-1}(\sqrt{d}))$$

A standard Gaussian in high dimensions behaves like a uniform sphere distribution

KL-Induced Posterior Collapse

Objective function of VAE:

$$\mathcal{L}_{\text{VAE}} = \mathbb{E}_{X\sim P_X}\Big[\mathbb{E}_{Z\sim Q_{Z|X}}[c(X, G(Z))] + \lambda\, D_{\text{KL}}(Q_{Z|X} \,\|\, P_Z)\Big]$$

ELBO-Surgery [2]:

$$\mathbb{E}_X[D_{\text{KL}}(Q_{Z|X}\|P_Z)] = I(X,Z) + D_{\text{KL}}(Q_Z\|P_Z)$$

$\downarrow D_{KL}(Q(z|x)\|P(z))$ may reduce the MI, leading to Posterior Collapse

From KL to Wasserstein

Objective function of WAE:

$$\mathcal{L}_{\text{WAE}} = \mathbb{E}_{P_X}[c(X, G(Z))] + \lambda \text{OT}(Q_Z, P_Z)$$

WAE replaces KL with Wasserstein between Q_Z (aggregated posterior) and P_Z

Spherical Sliced-Wasserstein [3]

$$\mathrm{SSW}_p^p(\mu,\nu) = \int_{S^{d-1}} W_p^p\left((\tilde{\mathcal{R}}\mu)^\theta, (\tilde{\mathcal{R}}\nu)^\theta\right) d\theta$$

$(\tilde{\mathcal{R}}\eta)^\theta$: spherical Radon transform

Hyperspherical Priors

❖ vMF:

$$\mathrm{vMF}(\mathbf{x};\mu,\kappa) = c_p(\kappa)\exp\left(\kappa\,\mu^\intercal\mathbf{x}\right),$$

$$\text{where } c_K(\kappa) = \frac{\kappa^{\frac{K}{2}-1}}{(2\pi)^{\frac{K}{2}} I_{\frac{K}{2}-1}(\kappa)}$$

❖ MvMF:

$$\mathrm{MvMF}(\mathbf{x};\boldsymbol{\Psi}) = \sum_{t=1}^{T}\alpha_t f_t(\mathbf{x}\mid\psi_t)$$

$$\psi_t = (\mu_t, \kappa_t)$$

$$\sum_{t=1}^{T}\alpha_t = 1, \quad \alpha_t \geq 0$$

$$i \sim \mathrm{Cat}(\alpha_1,\ldots,\alpha_T), \quad \mathbf{x} \sim f_i(\mathbf{x}\mid\psi_i)$$

❖ Uniform over S^{K-1}

$$\mathbf{x} \sim \mathcal{U}(\mathcal{S}^{K-1}) \Leftrightarrow \mathbf{x} = \frac{\mathbf{z}}{\|\mathbf{z}\|_2}, \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_K)$$

References:
[1] Joseph, et al. *Spherical Topic Models.* ICML, 2010.
[2] Hoffman et al. *ELBO Surgery: Yet Another Way to Carve Up the Variational Evidence Lower Bound.* AABI, NeurIPS Workshop, 2016.
[3] Bonet et al. *Spherical Sliced-Wasserstein.* ICLR, 2023.